

ORBITAL COLLISION ASSESSMENT

AEGIS ORBITAL — CLASSIFIED REPORT

Generated: Mon, 31 Aug 2026 05:26:27 GMT

Prediction Window: +49 hours

SATELLITE IDENTIFICATION

Name: ISS (ZARYA)
NORAD ID: 25544
Type: SPACE_STATION
Data Source: Aegis Orbital Cache (CACHED)
Status: OPERATIONAL

TWO-LINE ELEMENT (TLE) DATA

```
1 25544U 98067A 26242.62751382 .00005530 00000+0 10868-3 0 9998
2 25544 51.6315 291.6521 0005038 90.5345 269.6220 15.48940460583296
```

CONJUNCTION ANALYSIS

1 potential conjunction event(s) detected within 49-hour window

#	DEBRIS / OBJECT	OBJECT ID	TIME OF CLOSEST APPROACH (UTC)	MISS DIST. (km)	REL. SPEED (km/s)	RISK
1	FENGYUN 1C DEB	31113	2026-08-31 06:07:22 UTC	0.380	12.10	HIGH

RISK ASSESSMENT & METHODOLOGY

Aegis Orbital determines potential close approaches by propagating the primary and secondary objects using **SGP4** and calculating their relative separation over the prediction window. For each candidate pair: $\mathbf{r_rel} = \mathbf{r_secondary} - \mathbf{r_primary}$, miss distance $d = ||\mathbf{r_rel}||$, and relative velocity $\mathbf{v_rel} = ||\mathbf{v_secondary} - \mathbf{v_primary}||$. The system identifies the minimum predicted separation and corresponding Time of Closest Approach (TCA).

Standard TLE data does not contain complete orbital covariance information. Therefore, this report provides a simplified conjunction-risk assessment rather than an operational probability of collision. Professional collision probability assessment requires higher-fidelity state information and uncertainty/covariance data.

ANALYSIS PARAMETERS & HIGHEST PRIORITY EVENT

ANALYSIS PARAMETERS	HIGHEST PRIORITY CONJUNCTION
Propagation Model: SGP4 Prediction Window: +49 hours Screening Threshold: 50.0 km Propagation Step: 60 seconds Coordinate System: TEME / ECEF Data Epoch: 2026-08-31 05:26:27 UTC	Primary: ISS (ZARYA) Secondary: FENGYUN 1C DEB TCA: 2026-08-31 06:07:22 UTC Miss Distance: 0.380 km Relative Speed: 12.10 km/s (Risk: HIGH) Priority is elevated due to the small predicted miss distance combined with high relative velocity.